
X-pedite M.D: Assistive Tooling for Efficient Pneumonia Triage via Multi-exit Diagnostics

Raj Appana *
New York University
sa9026@nyu.edu

Noam Cicurel †
New York University
nc2660@nyu.edu

Abstract

The integration of Artificial Intelligence (AI) into medical imaging has the potential to revolutionize humanity’s ability to preserve life, particularly in high volume environments like Emergency Rooms (ER). Pneumonia remains a leading cause of mortality globally, requiring rapid and accurate triage in emergency settings. While state of the art (SOTA) Deep Learning models like ConvNeXt achieve excellent performance on visual tasks, their base computational demands render them unsuitable for resource constrained embedded systems found in hospital radiology departments. This paper introduces **X-pedite M.D.**, an assistive optimized diagnostic framework designed to bridge the gap between high performance AI and legacy medical hardware to assist human experts. We propose a dynamic inference architecture utilizing **Early Exits** and **Post Training Quantization (PTQ)** to reduce model size by 75% (from 748MB to 187MB) and accelerate inference latency by 2.69x (4.24ms vs 11.42ms). Our results demonstrate that heavy convolutional networks can be adapted for dynamic, real time, offline deployment and inference on edge devices with limited RAM (<512MB), thus providing a scalable solution for AI assisted triage in low resource environments.

1 Introduction

Multiple studies and statistics show that Pneumonia constitutes a persistent global health crisis, accounting for an unfortunately large number of deaths globally (2.5 million (2023))(1) and over 1 million hospitalizations annually just within the US(2). The implication of this is severe operational strain on hospital infrastructure. High volume patient intakes can very easily lead to critical bottlenecks in hospital triage queues, which is further exacerbated during large scale respiratory related pandemics such as COVID-19.

Furthermore, the human cost of this operational saturation should also be considered. The sustained cognitive load placed on Radiologists during these times may result in decision fatigue; a well documented human phenomenon which degrades diagnostic accuracy over time. As infrastructure bottlenecks delay patient care, the concurrent exhaustion of medical personnel increases the probability of diagnostic error which results in further patient care cases requiring triage and assessment. It is imperative to break this feedback loop and allow medical infrastructure and personnel to perform to their best ability (3).

To this end we explore assistive AI solutions. A significant disparity exists between the computational requirements of modern State of the art (SOTA) models and the hardware capabilities of medical edge devices. Mobile X-ray carts and portable Digital Radiography (DR) panels often rely on embedded processors with strict power and memory constraints for reliability and security reasons, meaning allocatable disk and RAM space for a third party application is very low. Additionally, while recent cutting edge automatic diagnostic tools use very large and powerful SOTA models, such approaches

*Courant Institute of Mathematical Sciences

†Tandon School of Engineering

introduce complex ethical, legal, and logistical liabilities. In contrast, we re frame the objective to assistive methodologies which help human experts as opposed to trying to do their job.

Architectures such as ConvNeXt-Large require around 800MB of storage, with later SOTA models demanding far more. This introduces a need for online connectivity and federated learning techniques. This reliance on connectivity introduces latency, privacy risks (GDPR compliance), additional computational and communication overhead, and operational points of failure in rural or disaster stricken areas. To address this, we present **X-pedite M.D.**, a streamlined assistive inference engine that enables offline, real time Pneumonia triage. We create this by modifying a pretrained ConvNeXt backbone with intermediate "Early Exit" classifiers and applying aggressive quantization to achieve a system capable of high confidence image predictions in lightning fast speeds (4 milliseconds), ensuring critical cases are flagged for immediate radiologist review without overwhelming the device hardware and alleviating a Radiologist's mental workload, thus delaying decision fatigue.

2 Individual Contribution

Note: All sections of this project were worked on collaboratively, sections outlined below denote which had more focus from each member:

Raj: Environment setup, Model load, Fine tuning, Early exit implementation, Ablation analysis, Feasibility analysis

Noam: Problem domain research, Quantization implementation, Development environment tooling, Evaluations, Hardware research, Dataset pre-processing

3 Problem Definition

The deployment of assistive triage systems in medical environments currently face two significant barriers: the logistical inefficiency of end to end sequential diagnostics and processing within the hardware limitations of medical equipment.

3.1 Triage Bottleneck

Assuming a standard patient triage flow, we observe the following steps: Patient A enters ER and awaits triage, Preliminary screening and emergency diagnostics performed by nurses, X-ray screens wait for review on a Picture Archiving and Communication System (PACS), Radiologist acquires Patient A screens and undertakes full assessment (with 0 immediate knowledge about the case pre assessment), Patient A is given prescription and then released.

This flow suffers from significant temporal inefficiencies, primarily due to the passive nature of PACS. Upon acquisition, X-ray scans remain static on the server and are queued sequentially until radiologist availability permits review. While manual triage protocols exist, they rely on ad hoc human assessment, introducing variability and delay. Furthermore, when a radiologist eventually initiates review, they do so "cold" i.e. without preliminary analysis or any form of visual attention guidance.

Consequently, patients with time sensitive conditions, such as severe pneumonia, are forced to endure damaging wait times. In mass casualty scenarios or pandemic surges (e.g. COVID-19), minimizing this diagnostic "blind spot" is critical. By implementing dynamic, confidence based reordering at the time the scan is taken, we can systematically prioritize high probability positive cases, thus reducing diagnostic turnover time for critical patients and optimizing the allocation of scarce human medical expertise.

3.2 Model Scaling

Computational availability in medical imaging is affected by severe resource constraints due to tight security specifications. Unlike general purpose computers, medical diagnostic equipment, such as DR panels and mobile X-ray carts, are required to last a long time, often remaining in service for over a decade after being provisioned due to rigorous regulatory certification processes. Once certified, the hardware specifications are effectively "frozen" to ensure consistent performance over this extended duration of time.

Consequently, these embedded systems are engineered with minimal overhead, strictly provisioning power and memory solely for image acquisition and basic signal processing. This leaves minimal capacity for third party software. In practice, the computational budget available for onboard machine learning inference is often only about 512MB. This limitation presents a fundamental incompatibility with applying modern Deep Learning and ViT models; SOTA architectures typically require gigabytes of storage and substantial throughput. Deploying these models on legacy hardware will inevitably lead to system instability, ranging from out of memory (OOM) issues all the way to possible data corruption, both of which are unacceptable for critical equipment.

3.3 The need for Offline Solutions

Conventional solutions to mitigate local resource constraints typically involve offloading computational burdens to cloud based infrastructure and bespoke forms of federated learning. While effective in a high connectivity environment, this dependency renders model deployment unfeasible in rural, remote, or bandwidth limited areas where stable network access cannot be guaranteed. Additionally, the transmission of sensitive patient data across networks to remote cloud servers introduces significant latency penalties and privacy liabilities, complicating compliance with rigorous data protection laws such as GDPR.

These problems necessitate a strictly defined solution space: the model architecture of an assistive tool must be engineered to operate within the residual memory of legacy embedded systems without significantly compromising the diagnostic accuracy and speed required for medical triage flows.

4 Related Work

4.1 Models in Medical Imaging

Recent advancements in medical computer vision have been dominated with models performing visual segmentation tasks. Notable among these is **SAM-Med2D** (4), which applies the Segment Anything Model (SAM) on a massive corpus of 4.6 million medical images. While such architectures demonstrate amazing segmentation capabilities, the authors acknowledge difficulty in training the model to segment medical images due to the ambiguous and convoluted nature of the data. To combat this, the authors explore the integration of specialized ground truth masks within the training pipeline. This strategy yields significant performance enhancements. These findings also provide empirical validation for the utility of assistive methodologies, suggesting that AI is most effective when integrated into a medical professional's workflow, rather than to automate it altogether.

4.2 Barriers to Clinical Deployment

Despite the use of high performance algorithms, the application of AI remains a significant challenge. A recent joint consensus statement by experts from the *RSNA and MICCAI* societies (5) identifies clinical integration and computational feasibility as primary bottlenecks preventing widespread adoption. The report emphasizes that for AI to be practically viable, it must operate within existing hospital infrastructures rather than requiring bespoke, high cost computing environments. **X-pedite M.D.** directly addresses this recommendation by targeting the medical compute boundary of legacy devices, ensuring that advanced diagnostics can be deployed without necessitating hardware overhauls.

4.3 AI as an Assistive Tool

The philosophy of this project entails AI as an assistive tool rather than an autonomous replacement of human expertise. This is supported by broader trends. A 2025 cross sectional study published in the *Journal of Medical Internet Research (JMIR)* surveyed over 3,000 global researchers and identified "physician workload reduction" and "decision fatigue mitigation" as the most critical value drivers for integrated AI in the medical field (6). The study concludes that the next generation of diagnostic tools should focus on streamlining workflows and acting as a "second opinion" to safeguard human expert acuity. By designing our system to dynamically reorder triage queues based on risk and provide classification confidence information to Radiologists, we align with this objective very tightly, positioning our model not as a replacement for the radiologist, but as a mechanism to optimize their productivity during high volume crises and day to day work.

5 Method

Our proposed framework, X-pedite M.D., integrates architectural adaptations with hardware optimizations to enable high performance inference on constrained edge devices. The pipeline consists mainly of the three stages below.

5.1 Data Analysis, Processing and Streaming

We utilized a carefully curated dataset of Chest X-Ray Images (Pneumonia labels), comprising 6,464 X-ray scan images labeled as either *Pneumonia* or *Normal*. The dataset underwent rigorous quality control, including manual verification by multiple expert physicians who manually pruned low quality or unreadable scans, ensuring high clinical fidelity. You may note an inherent skew in the number of pneumonia labels as opposed to normal ones, this was intentionally maintained to encourage false positive skewed "paranoia" into our model. In medical classification false negatives can have very dangerous and life threatening consequences. To mitigate this, the data skew was maintained to inherently encourage the model to be "paranoid", selecting cases where it was sure the scan it was classifying did not have pneumonia. The idea is to be "better safe than sorry" and classify predictions it's doubtful of as pneumonia which can then be flagged for detailed review by a Radiologist. To align the grayscale scans with the architectural requirements of the ConvNeXt backbone, we implemented a Pseudo RGB projection strategy where we replicated the single color band (black/white) across 3 channels. This was done to maintain tensor compatibility while preserving all semantic content in the image.

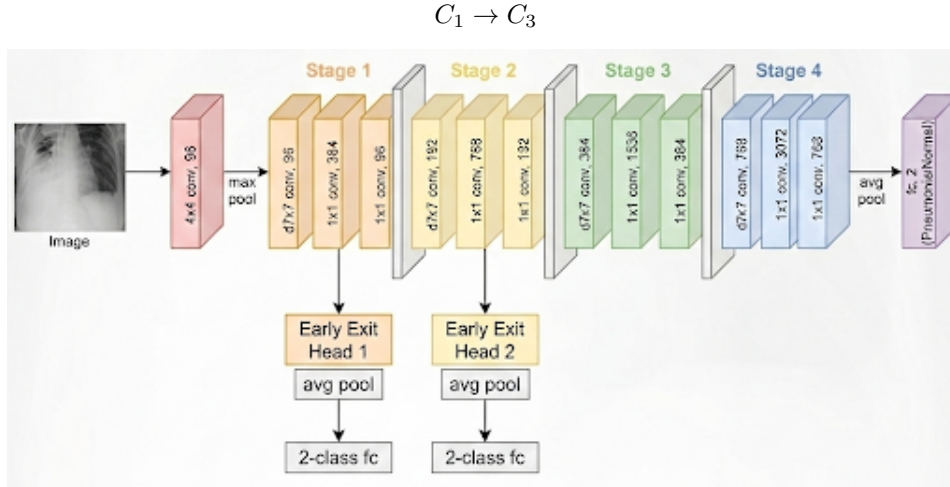


Figure 1: **X-pedite M.D. Architecture** The input chest X-ray flows through the ConvNeXt backbone. Two early exit classifiers are attached at Stage 1 and Stage 2, allowing high confidence predictions to terminate inference early. The final classifier head processes ambiguous samples that traverse the full length of the model.

5.2 ConvNeXt

ConvNeXt was chosen for our models backbone because it is a modern convolutional neural network (CNN) architecture that challenges more complex vision transformer (ViT) models yet rivals their performance, achieving 87.8% accuracy on ImageNet (7), all while maintaining the speed and simplicity that CNN models have over ViT models. ConvNeXt's crowning architectural improvements are it's inverse bottleneck residual links and large kernels. These enable greater global context capture and fine feature understanding from images respectively, both of which are critical for our domain.

5.3 Early Exits

We augmented the ConvNeXt backbone with two early exit classification heads, positioned in Stage 1 and Stage 2. This placement was selected based on the observation that computational savings diminish significantly at deeper layers; exiting at Stage 3 or 4 offers much less latency benefits relative to earlier stages. Topologically, each exit consists of a softmax and average pooling layer followed by a lightweight 2 channel fully connected classifier head.

This facilitates dynamic inference: if the prediction confidence at Exit 1 exceeds a calibrated hyperparameter threshold (τ), inference terminates immediately, handling "obvious" samples with minimal computation. Exit 2 acts as an intermediate filter for samples requiring more feature extraction and complex understanding, yet still effectively pruning the remaining 50% of the network. Finally, the "normal" classification head (modified from the original 1000 channels) is invoked solely for high ambiguity cases where the model requires the full semantic depth of the network to classify the scan. Exit conditions were determined using a softmax function output compared to an arbitrary threshold: $P(\hat{y}_i) > \tau$

To maintain high model performance, we utilized a targeted freezing optimization strategy. We trained only the exit heads (rather than updating the entire parameter space) by stopping gradient flow for ConvNeXt during the backward pass, effectively freezing the model backbone. This isolation significantly accelerated convergence and reduced training latency, allowing our weighted joint loss function to efficiently optimize decision boundaries of the exits without destabilizing the ConvNeXt network.

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{CE}(\hat{y}_1, y) + \lambda_2 \mathcal{L}_{CE}(\hat{y}_2, y) + \lambda_{final} \mathcal{L}_{CE}(\hat{y}_{final}, y)$$

or more simply

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{Exit1} + \lambda_2 \mathcal{L}_{Exit2} + \lambda_{final} \mathcal{L}_{Final}$$

Utilizing this combined loss function allowed effective tuning of the exit heads, we assign higher importance to the final output because the classifications going through the whole pipeline are ambiguous cases and thus much more "important" when applied to clinical context as opposed to the more confident predictions from the early exit outputs.

This joint loss function enabled the cohesive and parallel optimization of the early exits and final classifier. Crucially, we implemented a skewed weighting where the final classifier is assigned a higher scalar coefficient. This configuration reflects a deliberate prioritization of clinical safety, samples that traverse the full depth of the network typically represent the most semantically ambiguous and therefore diagnostically critical instances. By up weighting the final loss component, we enforce stricter penalties for errors on these complex cases, ensuring that the model's deepest (and most expensive) classifier head maintains maximum acuity for the patients who need it most.

5.4 Post training Quantization (PTQ)

To assess the feasibility of deploying our architecture on resource constrained embedded medical hardware, we conducted a series of Post Training Quantization (PTQ) experiments aimed at minimizing memory footprint without compromising diagnostic accuracy. We systematically compressed the baseline (FP32) weights into FP16 and INT8 formats. To maintain the semantic fidelity of the learned feature representations, we used a quantization scheme bounded by scale factors and dynamically calibrated clipping ranges.

Furthermore, to ensure numerical stability during low precision arithmetic, we implemented a clamping mechanism to handle vanishing gradients and division by zero edge cases. By imposing a fixed offset on near zero values, we effectively prevented the propagation of catastrophic numerical errors through our network, ensuring consistent inference behavior even at high compression ratios.

6 Experiment Results

6.1 Ablation Study

To quantify the individual contributions of our optimization components, we conducted a systematic ablation study. The results isolate the distinct roles of Post Training Quantization (PTQ) and the multi exit architecture in addressing the dual constraints of memory and speed. Table 1 reflects these results.

While quantization yielded only marginal reductions in inference latency, its primary contribution lies in the significant compression of the model's memory, as illustrated in Figure 3. Both the FP16 and INT8 configurations successfully undershoot the strict 512MB memory ceiling of legacy radiography infrastructure available for third party tools. This compression also preserves essential memory

headroom, ensuring that sufficient RAM remains available to accommodate the dynamic tensor computations and pre processing overhead required during real time inference.

By successfully offloading confident classifications at the first and second exits, the system achieves an average inference latency of **4.63ms**, representing a **2.47x speedup** compared to the 11.42ms baseline of the full precision model. Figures 2 and 4 clearly illustrate this speedup and the tunable trade off between the confidence threshold and system throughput; as the threshold increases, the system forces more scans to flow through more parts of the network, thus increasing accuracy while linearly degrading speedup.

Figure 2: **Accuracy/Latency results across different model configurations.**

Configuration	Acc (%)	Lat (ms)
FP32 (Base)	84.26	11.42
FP32 + 0.9 Exit	79.97	5.04
FP32 + 0.8 Exit	81.34	4.23
FP16	83.94	11.23
FP16 + 0.9 Exit	79.97	5.05
FP16 + 0.8 Exit	81.13	4.25
INT8	86.41	11.07
INT8 + 0.9 Exit	80.11	4.96
INT8 + 0.8 Exit	79.97	4.24

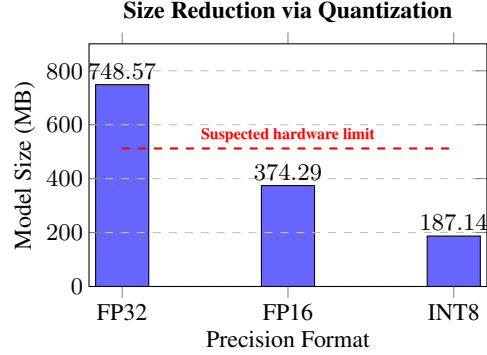


Figure 3: **Quantized Model sizes**

An analysis of the exit distributions during validation reveals that the vast majority of samples are successfully resolved at the shallowest layers of the network. Approximately **80%** of the test set exited at Stage 1, insinuating that a significant portion of scans contain structurally obvious features that do not require deep semantic processing. This distribution (Figure 5) validates the efficiency of our dynamic architecture: the heavy computational cost of the model is reserved for the minority of scans which are difficult to classify.

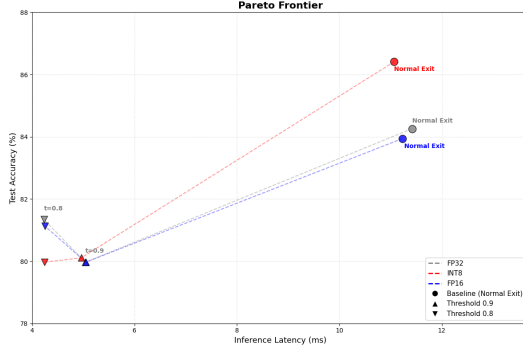


Figure 4: **Ablations visualizations**

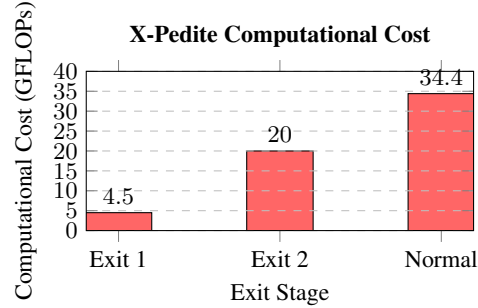


Figure 5: **Computational savings**

6.2 Classifier Behavior and Error Analysis

The Grad CAM analysis demonstrates that the model correctly attends to clinically relevant features. As shown in Figure 7, the activation hotspots consistently align with variable opacity regions and detailed pulmonary structures in the lungs. This precise localization confirms that the model is effectively learning the structural biomarkers of pneumonia. The general blue hue of the image also demonstrates the consideration of global context for output classification, an essential detail when handling X-ray images.

To assess the model's reliability in applied clinical use cases we assessed the model's sensitivities using a confusion matrix. Via Figure 6, quantitative analysis of the confusion matrix confirms the success of our class imbalance strategy. The model exhibits a distinct **"Paranoid"** behavior (deliberate bias

towards high sensitivity (Recall)). The rate of false negatives are minimized due to this, indicating that the system prioritizes safe classifications (False Positives). This behavior is a desirable safety feature, ensuring that potential pneumonia classifications are flagged for human review rather than being silently dismissed as normal.

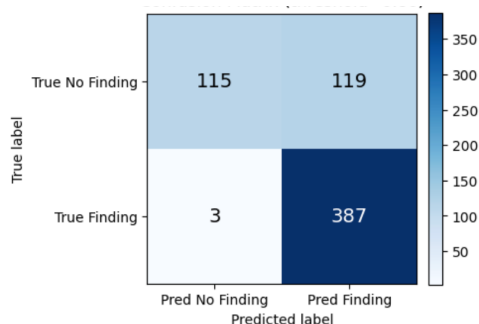


Figure 6: Confusion Matrix

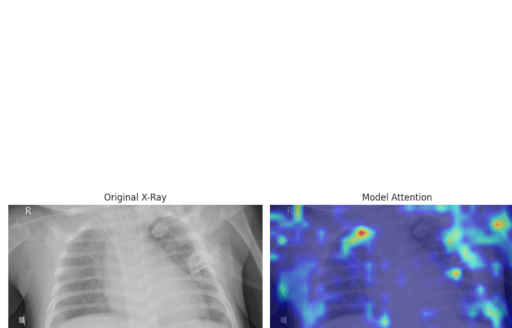


Figure 7: Attention Heatmap

7 Conclusion

This study demonstrates that assistive AI tooling need not be tethered to prohibitively expensive GPU workstations while also maintaining high performance when applied to tasks that assist medical professionals. By synergizing architectural adaptation (Early Exits) with hardware optimizations (Post Training Quantization), **X-pedite M.D.** successfully bridges the technical gap and bottleneck in the pneumonia triage flow. We achieve a 75% reduction in model size (187MB) and a 2.47x acceleration in inference latency (4.24ms), rendering the system fully compatible with the expected 512MB memory constraints of legacy digital radiography panels and X-ray carts. This satisfies coverage of the earlier technical bottlenecks identified in the patient triage flow and thus would be beneficial to Radiologists in the field.

Most importantly, this efficiency was not achieved at the expense of patient safety. Our training strategy and dynamic joint loss function yielded a model with high recall sensitivity, adhering to strict medical standards by minimizing false negatives. By designing the system to act as a triage partner (clearing obvious cases quickly while flagging ambiguous ones for expert review) we validate an assistive workflow that aids, rather than replaces, Radiologists.

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